

managing projects. The schedule is the essential tool for managing the project according to the project work plan.

Scheduling software is used to perform the many calculations in a CPM schedule. Scheduling software is good for quickly performing calculations for early/late start and finish times of activities, total and free float times, and cost distributions on an early, late, and target basis. However, scheduling software cannot identify activities and cannot determine the logical sequencing of activities. Only people can identify and sequence work activities. Therefore, entering the project plan data into the scheduling software should be the last activity of project planning.

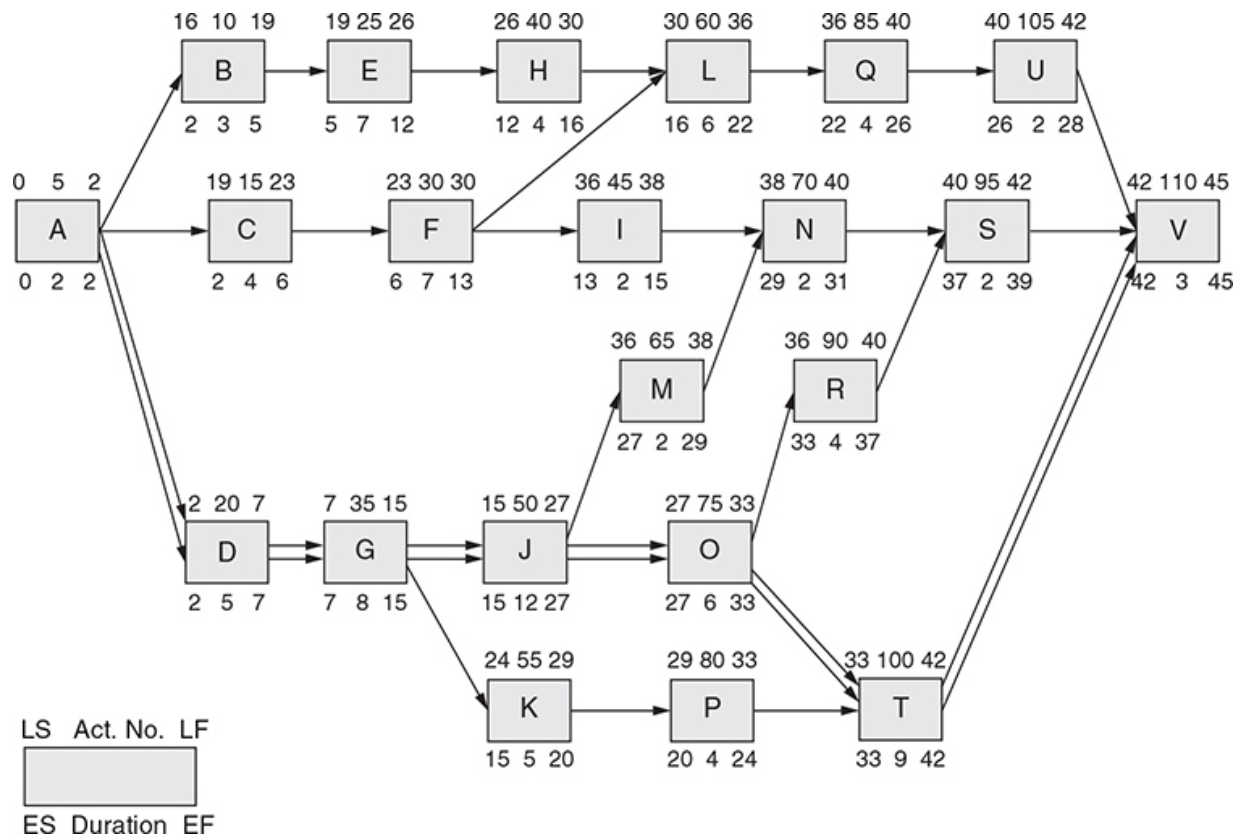
Before entering data into a scheduling software package, a careful review of the list of activities should be performed to ensure that work is sufficiently defined to complete the project. A logic diagram should be created and reviewed to ensure that activities are sequenced in a realistic order to guide field operations. There should also be a careful check of the duration that is assigned to each activity to ensure the work can be accomplished in a reasonable time. These items can only be accomplished by experienced people, not the software.

It is easy for people to become complacent and depend too much on computer analysis. Although the computer can perform calculations in seconds, it is highly likely that errors are created in the process of entering data in the computer. Therefore, it is necessary to check the output from the computer to ensure a quality schedule. Example 10-1 shows hand calculations that are easily performed to check and verify the results of computer analysis of a cost-loaded CPM diagram. For example, it is easy to perform a forward pass on the CPM diagram to calculate the early start and finish of an activity as a check to verify the correct output from the computer.

Example 10-1 A project team has identified the following activities required to complete a project. The duration of each activity is based on the quantity of work, available resources, and productivity of crews. The cost of each activity is based on quantity of work and historical cost data from previously completed projects.

Activity	Activity No.	Duration (days)	Cost	Predecessor(s)	Successor(s)
A	5	2	\$500	None	B, C, D
B	10	3	\$900	A	E
C	15	4	\$1,600	A	F
D	20	5	\$500	A	G
E	25	7	\$1,400	B	H
F	30	7	\$1,500	C	I, L
G	35	8	\$2,400	D	J, K
H	40	4	\$800	E	L
I	45	2	\$1,000	F	N
J	50	12	\$3,600	G	M, O
K	55	5	\$2,000	G	P
L	60	6	\$1,200	F, H	Q
M	65	2	\$900	J	N
N	70	2	\$700	I, M	S
O	75	6	\$1,800	J	R, T
P	80	4	\$1,200	K	T
Q	85	4	\$2,000	L	U
R	90	4	\$1,600	O	S
S	95	2	\$1,400	N, R	V
T	100	9	\$1,800	O, P	V
U	105	2	\$1,200	Q	V
V	110	3	\$300	S, T, U	None

From the preceding information, a CPM precedence diagram can be developed to calculate the schedule for the project. The critical path is shown as double arrows between critical activities.



Example 10-1 Precedence diagram using data table above. (All time calculations are based on end of working day.)

From the calculations shown on the CPM diagram, the following table is prepared, which shows the early/late starts and finishes as well as the total and free float for each activity. All times are end of workday. For example, Activity A starts at the end-of-day 0, which is the beginning of day 1.

The early start is the earliest time an activity can be started. It is governed by the largest early finish of the immediately preceding activity. The early finish is the earliest time an activity can be finished and is equal to the early start plus the duration of the activity. The late start is the latest time an activity can be started without delaying the completion date of the project. It is calculated as the late finish minus the duration of the activity. The late finish is the latest time an activity can be finished without delaying the completion date of the project. It is governed by the smallest late start of the immediately following activity.

The total float is the amount of time an activity may be delayed without delaying the completion date of the project. It is calculated as the late finish minus the early finish, or as the late start minus the early start of the activity. The free float is the amount of time an activity may be delayed without delaying the early start time of the immediately following activity. The free float of an activity is calculated as the early start of the following activity minus the early finish of the activity. Critical activities are those activities that have a total float and free float equal to zero. The critical path activities are noted as C in the right-hand column of the following table.

Activity no.	Activity	Pred.(s)	Succ.(s)	Duration, days	Total cost	Note: All time calculations are based on end of working day						Critical
						ES	EF	LS	LF	TF	FF	
5	A	None	B, C, D	2	\$500	0	2	0	2	0	0	C
10	B	A	E	3	\$900	2	5	16	19	14	0	
15	C	A	F	4	\$1,600	2	6	19	23	17	0	
20	D	A	G	5	\$500	2	7	2	7	0	0	C
25	E	B	H	7	\$1,400	5	12	19	26	14	0	
30	F	C	I, L	7	\$1,500	6	13	23	30	17	0	
35	G	D	J, K	8	\$2,400	7	15	7	15	0	0	C
40	H	E	L	4	\$800	12	16	26	30	14	0	
45	I	F	N	2	\$1,000	13	15	36	38	23	14	
50	J	G	M, O	12	\$3,600	15	27	15	27	0	0	C
55	K	G	P	5	\$2,000	15	20	24	29	9	0	
60	L	F, H	Q	6	\$1,200	16	22	30	36	14	0	
65	M	J	N	2	\$900	27	29	36	38	9	0	
70	N	I, M	S	2	\$700	29	31	38	40	9	6	
75	O	J	R, T	6	\$1,800	27	33	27	33	0	0	C
80	P	K	T	4	\$1,200	20	24	29	33	9	9	
85	Q	L	U	4	\$2,000	22	26	36	40	14	0	
90	R	O	S	4	\$1,600	33	37	36	40	3	0	
95	S	N, R	V	2	\$1,400	37	39	40	42	3	3	
100	T	O, P	V	9	\$1,800	33	42	33	42	0	0	C
105	U	Q	V	2	\$1,200	26	28	40	42	14	14	
110	V	S, T, U	None	3	\$300	42	45	42	45	0	0	C

Management is always interested in both time and cost. The cost of each activity is distributed over the duration of the activity. However, some activities may begin on their early start date, whereas other activities may begin on their late start date. Likewise, some activities may finish on their early finish date, whereas other activities may finish on their late finish dates. Therefore, costs may be analyzed on an early start basis, or a late start basis. Following is a cost analysis of the project on an early start and late start basis. The S-curve for the project shows the cost/time relationship with percent costs and percent times.

Early Start Cost Analysis (Note Activity F cost of \$214.29 is used as \$214.00 on days 7 through 13. The rounding is adjusted in the early start cumulative cost on Day 13.)

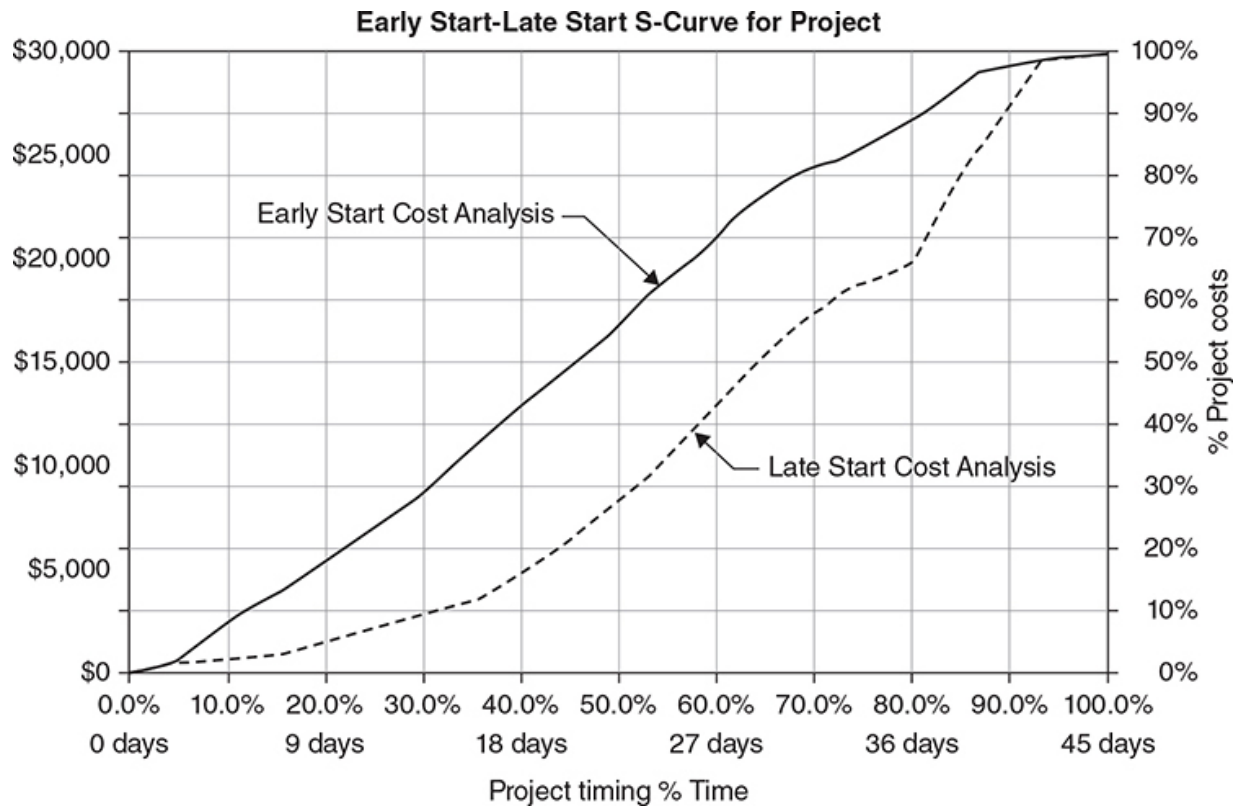
Day	ES % time	Early start basis activities in progress	ES \$/day	ES Cum. \$	ES % Cost
1	2.2%	A@\$250	\$250	\$250	0.8%
2	4.4%		\$250	\$500	1.7%
3	6.7%		\$800	\$1,300	4.3%

4	8.9%	B@\$300 + C@\$400 + D@\$100	\$800	\$2,100	6.9%
5	11.1%		\$800	\$2,900	9.6%
6	13.3%	C@\$400 + D@\$100 + E@\$200	\$700	\$3,600	11.9%
7	15.6%	D@\$100 + E@\$200 + F@\$214	\$514	\$4,114	13.6%
8	17.8%	E@\$200 + F@\$214 + G@\$300	\$714	\$4,828	15.9%
9	20.0%		\$714	\$5,542	18.3%
10	22.2%		\$714	\$6,256	20.6%
11	24.4%		\$714	\$6,970	23.0%
12	26.7%		\$714	\$7,684	25.4%
13	28.9%	F@\$214 + G@\$300 + H@\$200	\$714	\$8,400	27.7%
14	31.1%	G@\$300 + H@\$200 + I@\$500	\$1,000	\$9,400	31.0%
15	33.3%		\$1,000	\$10,400	34.3%
16	35.6%	H@\$200 + J@\$300 + K@\$400	\$900	\$11,300	37.3%
17	37.8%	J@\$300 + K@\$400 + L@\$200	\$900	\$12,200	40.3%
18	40.0%		\$900	\$13,100	43.2%
19	42.2%		\$900	\$14,000	46.2%
20	44.4%		\$900	\$14,900	49.2%
21	46.7%	J@\$300 + L@\$200 + P@\$300	\$800	\$15,700	51.8%
22	48.9%		\$800	\$16,500	54.5%
23	51.1%	J@\$300 + P@\$300 + Q@\$500	\$1,100	\$17,600	58.1%
24	53.3%		\$1,100	\$18,700	61.7%
25	55.6%	J@\$300 + Q@\$500	\$800	\$19,500	64.4%
26	57.8%		\$800	\$20,300	67.0%
27	60.0%	J@\$300 + U@\$600	\$900	\$21,200	70.0%
28	62.2%	M@\$450 + O@\$300 + U@\$600	\$1,350	\$22,550	74.4%
29	64.4%	M@\$450 + O@\$300	\$750	\$23,300	76.9%
30	66.7%	N@\$350 + O@\$300	\$650	\$23,950	79.0%
31	68.9%		\$650	\$24,600	81.2%
32	71.1%	O@\$300	\$300	\$24,900	82.2%
33	73.3%		\$300	\$25,200	83.2%
34	75.6%	R@\$400 + T@\$200	\$600	\$25,800	85.1%
35	77.8%		\$600	\$26,400	87.1%
36	80.0%		\$600	\$27,000	89.1%
37	82.2%		\$600	\$27,600	91.1%
38	84.4%	S@\$700 + T@\$200	\$900	\$28,500	94.1%
39	86.7%		\$900	\$29,400	97.0%
40	88.9%	T@\$200	\$200	\$29,600	97.7%
41	91.1%		\$200	\$29,800	98.3%
42	93.3%		\$200	\$30,000	99.0%
43	95.6%	V@\$100	\$100	\$30,100	99.3%
44	97.8%		\$100	\$30,200	99.7%
45	100.0%		\$100	\$30,300	100.0%

Late Start Cost Analysis (Note Activity F daily cost of \$214.29 is rounded to \$214 on days 24 through 30. The rounding is adjusted in the late start cumulative cost on Day 30.)

Day	LS % time	Late start basis activities in progress	LS \$/day	LS Cum. \$	LS % Cost
1	2.2%	A@\$250	\$250	\$250	0.8%
2	4.4%		\$250	\$500	1.7%
3	6.7%	D@\$100	\$100	\$600	2.0%
4	8.9%		\$100	\$700	2.3%
5	11.1%		\$100	\$800	2.6%
6	13.3%		\$100	\$900	3.0%
7	15.6%	G@\$300	\$100	\$1,000	3.3%
8	17.8%		\$300	\$1,300	4.3%
9	20.0%		\$300	\$1,600	5.3%
10	22.2%		\$300	\$1,900	6.3%
11	24.4%		\$300	\$2,200	7.3%
12	26.7%		\$300	\$2,500	8.3%
13	28.9%		\$300	\$2,800	9.2%
14	31.1%		\$300	\$3,100	10.2%
15	33.3%		\$300	\$3,400	11.2%
16	35.6%	J@\$300	\$300	\$3,700	12.2%
17	37.8%	B@\$300 + J@\$300	\$600	\$4,300	14.2%
18	40.0%		\$600	\$4,900	16.2%
19	42.2%		\$600	\$5,500	18.2%
20	44.4%	C@\$400 + E@\$200 + J@\$300	\$900	\$6,400	21.1%
21	46.7%		\$900	\$7,300	24.1%
22	48.9%		\$900	\$8,200	27.1%
23	51.1%		\$900	\$9,100	30.0%
24	53.3%	E@\$200 + F@\$214 + J@\$300	\$714	\$9,814	32.4%
25	55.6%	E@\$200 + F@\$214 + J@\$300 + K@\$400	\$1,114	\$10,928	36.1%
26	57.8%		\$1,114	\$12,042	39.7%
27	60.0%	F@\$214 + H@\$200 + J@\$300 + K@\$400	\$1,114	\$13,156	43.4%
28	62.2%	F@\$214 + H@\$200 + K@\$400 + O@\$300	\$1,114	\$14,270	47.1%
29	64.4%		\$1,114	\$15,384	50.8%
30	66.7%	F@\$214 + H@\$200 + O@\$300 + P@\$300	\$1,014	\$16,400	54.1%
31	68.9%	L@\$200 + O@\$300 + P@\$300	\$800	\$17,200	56.8%
32	71.1%		\$800	\$18,000	59.4%
33	73.3%		\$800	\$18,800	62.0%
34	75.6%	L@\$200 + T@\$200	\$400	\$19,200	63.4%
35	77.8%		\$400	\$19,600	64.7%
36	80.0%		\$400	\$20,000	66.0%
37	82.2%	I@\$500 + M@\$450 + Q@\$500 + R@\$400 + T@\$200	\$2,050	\$22,050	72.8%
38	84.4%		\$2,050	\$24,100	79.5%
39	86.7%	N@\$350 + Q@\$500 + R@\$400 + T@\$200	\$1,450	\$25,550	84.3%
40	88.9%		\$1,450	\$27,000	89.1%
41	91.1%	S@\$700 + T@\$200 + U@\$600	\$1,500	\$28,500	94.1%
42	93.3%		\$1,500	\$30,000	99.0%
43	95.6%	V@\$100	\$100	\$30,100	99.3%
44	97.8%		\$100	\$30,200	99.7%
45	100.0%		\$100	\$30,300	100.0%

S-Curve for the Project:



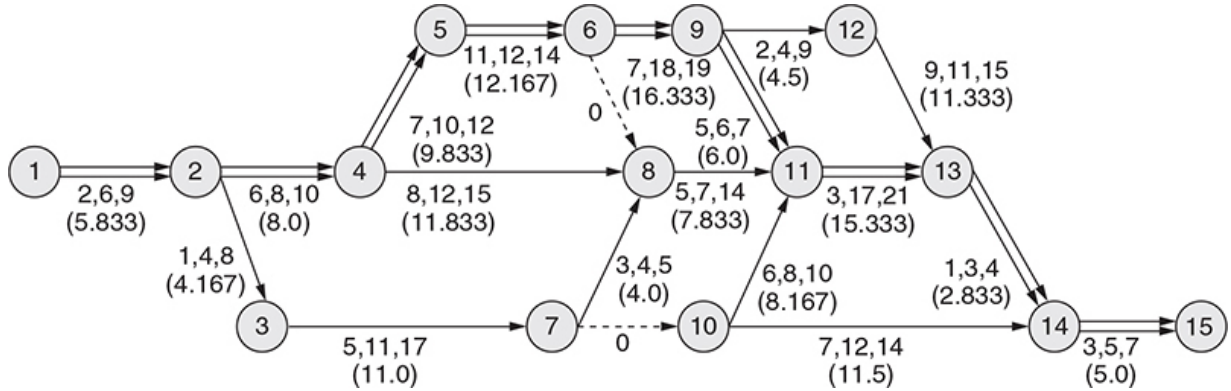
Program Evaluation and Review Technique

In the CPM of scheduling projects, the duration of each activity is usually defined with a reasonable degree of certainty. For most projects, the type and amount of work is known, which enables the project manager to establish the approximate duration for each work activity. For example, the time to produce drawings may be 4 weeks, the time to perform a soil investigation to gather and test soil samples may be 2 weeks, or the time to erect forms for concrete may be 3 days. Using the CPM, the assignment of one duration to each activity provides a deterministic process for the start and finish dates of each activity and a single finish date for the entire project.

For some projects, it may be difficult to estimate a reasonable single duration for one or more of the activities in the project schedule. There may be a range of durations that may apply to a particular activity, which makes it difficult to select just one duration to assign to the activity. The Program Evaluation and Review Technique (PERT) method of scheduling uses three

TABLE 10-8 PERT Probability Table

Example 10-2 The following PERT diagram shows the optimistic time (a), the most likely time (t_m), and the pessimistic time (b) at the bottom of each activity. The number shown in parentheses below each activity is the expected time (t_e) for the activity, calculated from the equation, $t_e = (a + 4m + b)/6$. All times denote working days in the project.



The duration of the project can be calculated by performing a forward pass to calculate the start and finish times using the optimistic duration (smallest duration) of each activity. For the data in this PERT diagram, the optimistic project duration is 48 days, which follows through events 1, 2, 4, 5, 6, 9, 12, 13, 14, and 15.

The pessimistic duration of the project can be calculated by performing a forward pass to calculate the start and finish times using the pessimistic (largest duration) of each activity. For the data in this PERT diagram, the pessimistic project duration is 103 days, which follows through events 1, 2, 4, 5, 6, 9, 11, 13, 14, and 15.

The project duration (T_e) for event 15 can be calculated by performing a forward pass to calculate the start and finish times using the expected duration (t_e) of each activity. For the data in this PERT diagram, the expected time of the last event number 15 is $T_e = 81.3$ days, which follows through events 1, 2, 4, 5, 6, 9, 11, 13, 14, and 15.

In summary, for this PERT diagram, the optimistic project duration is 48 days; the pessimistic project duration is 103 days; and the expected project duration is 81.3 days. The probability of completing the project ahead or behind day 81.3 will depend on the σ_{TE} of event 15. The value of σ_{TE} can be calculated as the square root of the sum of the variances of all activities along the critical path. Based on the expected time of each activity, the critical path for this PERT network follows events 1, 2, 4, 5, 6, 9, 11, 13, 14, and 15. The σ_{TE} for the final event, number 15, is the square root of the variances of these events, which can be calculated as follows:

$$\begin{aligned}
\sigma_{TE} &= \sqrt{v_{1-2} + v_{2-4} + v_{4-5} + v_{5-6} + v_{6-9} + v_{9-11} + v_{11-13} + v_{13-14} + v_{14-15}} \\
&= \sqrt{\left(\frac{9-2}{6}\right)^2 + \left(\frac{10-6}{6}\right)^2 + \left(\frac{12-7}{6}\right)^2 + \left(\frac{14-11}{6}\right)^2 + \left(\frac{19-7}{6}\right)^2 + \left(\frac{7-5}{6}\right)^2 + \left(\frac{21-3}{6}\right)^2 + \left(\frac{4-1}{6}\right)^2 + \left(\frac{7-3}{6}\right)^2} \\
&= \sqrt{1.3611 + 0.4444 + 0.6944 + 0.2500 + 4.0000 + 0.1111 + 9.0000 + 0.2500 + 0.4444} \\
&= \sqrt{16.5554} \\
&= 4.07
\end{aligned}$$

All the calculations necessary to perform an analysis of the project finishing ahead or behind the expected project duration of 81.3 days can be determined from the deviation equation. For example, it may be desired to determine the probability of the project finishing later than expected, that is, later than 81.3 days. The probability of completing the project by the 84th day can be calculated as follows:

$$\begin{aligned}
z &= \frac{T_S - T_E}{\sigma_{TE}} \\
&= \frac{84.0 - 81.3}{4.07} \\
&= 0.663
\end{aligned}$$

From [Table 10-8](#), based on a deviation z of 0.663, the probability is 0.75. Thus, there is a 75% probability that the project will be completed by the 84th day.

Other probability scenarios can be calculated. For example, the probability of the project finishing earlier than expected, such as in 78 days, can be calculated as follows:

$$\begin{aligned}
z &= \frac{T_S - T_E}{\sigma_{TE}} \\
&= \frac{78.0 - 81.3}{4.07} \\
&= -0.811
\end{aligned}$$

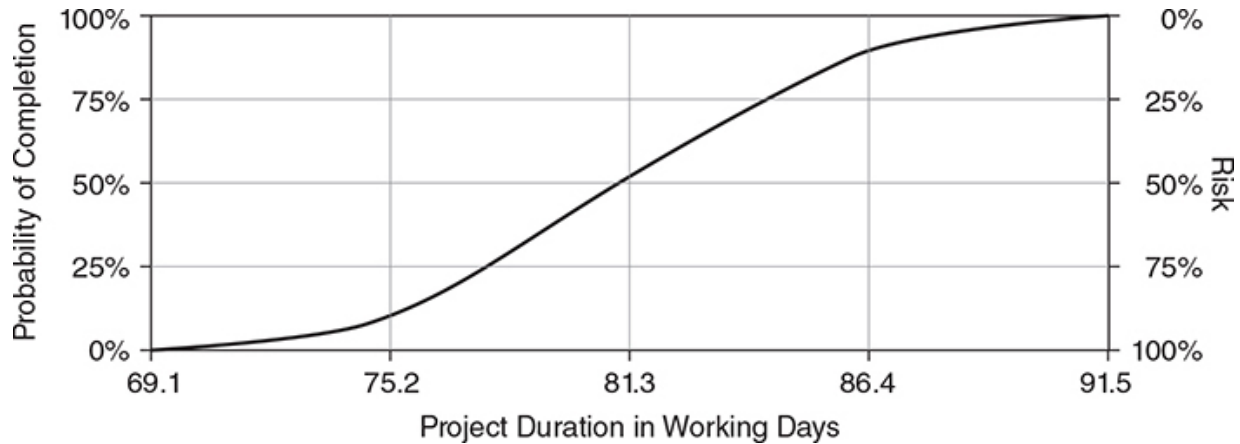
From [Table 10-8](#), based on a deviation z of -0.811 , the probability is 0.21. Thus, there is only a 21% probability that the project will be completed by the 78th day.

It may be desired to calculate the number of days to complete the project based on a 70% probability. This scenario can be calculated as follows. For a 70% probability, the value of z from [Table 10-8](#) is 0.5333:

$$\begin{aligned}
z &= \frac{T_S - T_E}{\sigma_{TE}} \\
0.533 &= \frac{T_S - 81.3}{4.07} \\
T_S &= 83.5 \text{ days}
\end{aligned}$$

Thus, there is a 70% probability that the project will be completed by 83.5 days.

The following is a graphical plot of the probability versus the project duration. This curve provides the full range of probabilities for assessing the risks of finishing the project ahead of or behind the expected duration of 81.3 days.



This analysis is based on a σ_{TE} for event 15 based on the critical path activities. A full analysis is more complex if all near-critical paths are analyzed. For example, it is possible to have another path through the PERT diagram that is near critical, which would yield a shorter T_e for event 15 and a higher value for σ_{TE} due to larger variances in the activities in the noncritical path.

Successor/Predecessor Relationships

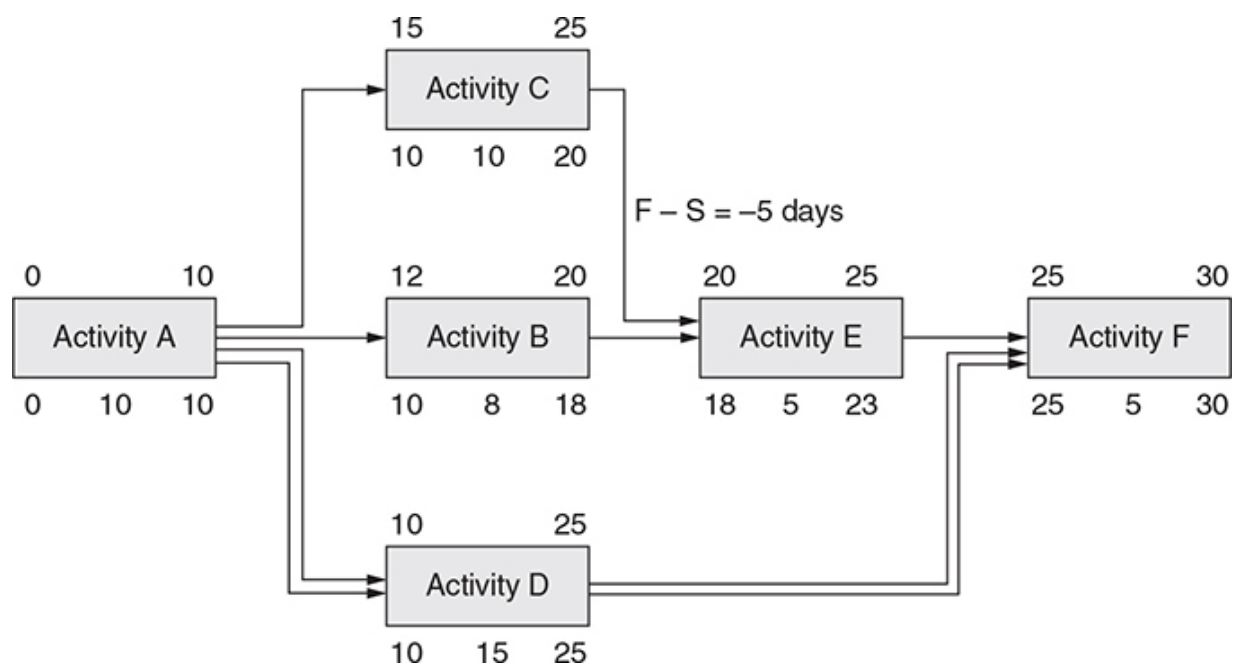
The CPM diagram is a graphical representation that shows the sequencing of activities in the project. The purpose of the CPM diagram is to model the logical flow of work, showing each activity and one or more activities that follow each activity. In the activity relationships, the preceding activity is called the predecessor activity, and the following activity is called the successor activity. In a pure CPM network diagram, a finish-to-start (F-S) relationship of each activity is assumed. That is, the preceding activity must be completed before starting the following activity. All the CPM diagrams and calculations presented earlier in this book are based on an F-S relationship of activities to depict a pure CPM logic network diagram.

Most commercially available computer software programs for CPM provide activity relationships other than F-S. These additional activity relationships include start-to-start (S-S), start-to-finish (S-F), and finish-to-finish (F-F). The S-S relationship means the successor activity can start at the same time or later than the predecessor activity. The F-F relationship means the successor activity can finish at the same time as or later than the

it is updated. If the schedule logic is finicky or requires special maintenance to maintain useful information with each progress update, it undermines its purpose. Also, the team and other stakeholders will lose confidence in the information and schedule predictions, thereby undermining the project manager's ability to manage the project to the stated objectives.

The following examples are presented to illustrate calculations based on each of the successor/predecessor activity relationships, F-S, S-S, F-F, and S-F. These examples show how each path must be carefully investigated to determine the start, finish, and float times. When the calculations are performed by computer software, it is possible to get project schedules that sometimes do not make logical sense. Thus, it is important to understand the calculations and be able to check the results of computer outputs to ensure the results are presented as intended by the person who prepared the schedule as well as by the persons who will be interpreting and using the schedule.

Example 10-3: Finish-to-Start Activity Relationships Calculate the start, finish, and float for each activity in the following CPM precedence network diagram. Activities C and E have an F-S relationship with a 5-day negative lag. The 5-day negative lag is equivalent to a 5-day lead, which means the finish of Activity C must occur no later than 5 days after the start of Activity E, given by the relationship: $EF_C \leq ES_E + 5$. The use of negative lags is not recommended because they lead to much confusion with respect to the meaning of the relationship and the schedule as discussed in the following paragraphs. The following diagram and accompanying table show the early and late start, early and late finish, and the total and free float for each activity.



LS	LF
Activity Description	
ES	EF

For Activity C, the calculation for total and free float are:

$$\begin{aligned}
 TF_C &= LF_C - EF_C \\
 &= 25 - 20 \\
 &= 5 \text{ days}
 \end{aligned}$$

$$\begin{aligned}
 FF_C &= ES_E - EF_C - \text{Lag} \\
 &= 18 - 20 - (-5) \\
 &= 3 \text{ days}
 \end{aligned}$$

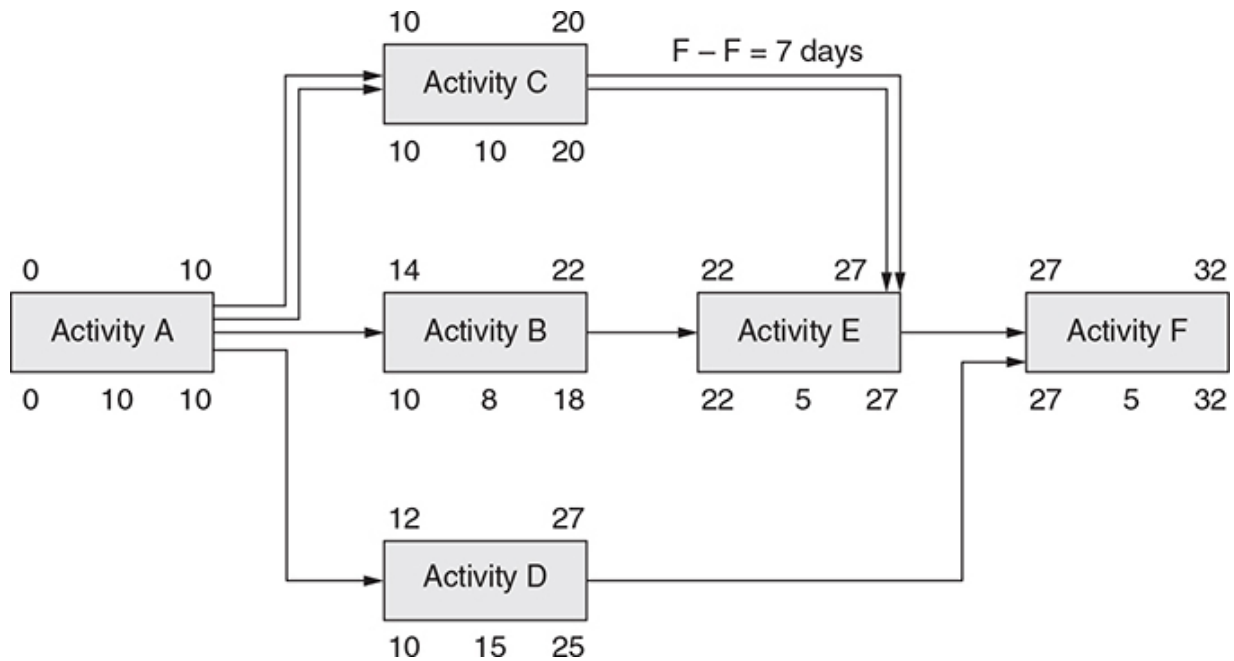
Activity	Duration	Early start	Early finish	Late start	Late finish	Total float	Free float
A*	10	0	10	0	10	0	0
B	8	10	18	12	20	2	0
C	10	10	20	15	25	5	3
D*	15	10	25	10	25	0	0
E	5	18	23	20	25	2	2
F*	5	25	30	25	30	0	0

*Denotes critical activities.

Based on the forward and backward pass calculations, the critical path is through Activities A, D, and F. Note the impact on the schedule and possible confusion due to the F-S relationship with 5 days of negative lag between Activities C and E. The EF_C is 20 days (governed by the equation $EF_C = ES_C + D = 10 + 10 = 20$). The EF_C is not governed by the equation $EF_C \leq ES_E + 5 = 18 + 5 = 23$ days, because the ES_E is governed by 18 from the EF_B . Also, a review of the diagram shows the preceding activity $EF_C = 20$, which is greater than the following activity $ES_E = 18$, which is illogical. A computer schedule that shows the ES of a following activity to be earlier than the EF of a preceding activity will make no sense to workers in the field and will cause them to lose confidence in the schedule.

Similar confusion exists for the free float of Activity C. The FF_C is governed by the equation $FF_C = ES_E - EF_C - \text{Lag} = 18 - 20 - (-5) = 3$ days. However, a review of the diagram shows the $EF_C = 20$, which is greater than the $ES_E = 18$. Thus, the use of F-S relationships with negative lag is not recommended.

Example 10-4: Finish-to-Finish Activity Relationships Calculate the start, finish, and float for each activity in the following CPM precedence network diagram. Activities C and E have an F-F relationship with a lag of 7 days. The following diagram and accompanying table show the early and late start, early and late finish, and the total and free float for each activity.



LS	LF
Activity Description	
ES	EF
D	

For Activity C, the calculation for total float and free float are:

$$\begin{aligned}
 TF_C &= LF_C - EF_C \\
 &= 20 - 20 \\
 &= 0 \text{ days}
 \end{aligned}$$

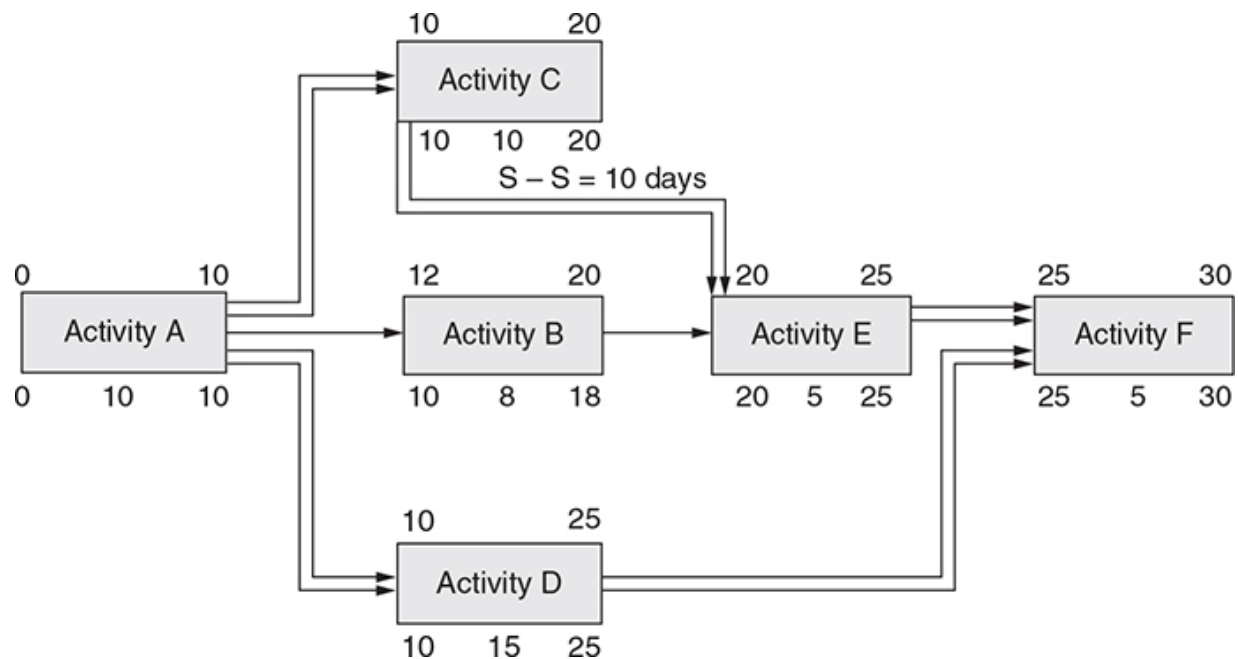
$$\begin{aligned}
 FF_C &= LF_E - LF_C - \text{Lag} \\
 &= 27 - 20 - 7 \\
 &= 0 \text{ days}
 \end{aligned}$$

Activity	Duration	Early start	Early finish	Late start	Late finish	Total float	Free float
A*	10	0	10	0	10	0	0
B	8	10	18	14	22	4	4
C*	10	10	20	10	20	0	0
D	15	10	25	12	27	2	2
E*	5	22	27	22	27	0	0
F*	5	27	32	27	32	0	0

*Denotes critical activities.

Based on the forward and backward pass calculations, the critical path is through Activities A, C, E, and F. Note that the free float of Activity C is governed by the F-F relationship.

Example 10-5: Start-to-Start Activity Relationships Calculate the start, finish, and float for each activity in the following CPM precedence network diagram. Activities C and E have an S-S relationship with a lag of 10 days. The following diagram and accompanying table show the early and late start, early and late finish, and the total and free float for each activity.



LS	LF
Activity Description	
ES	EF

For Activity C, the calculation for total float and free float are:

$$\begin{aligned}
 TF_C &= LF_C - EF_C \\
 &= 20 - 20 \\
 &= 0 \text{ days}
 \end{aligned}$$

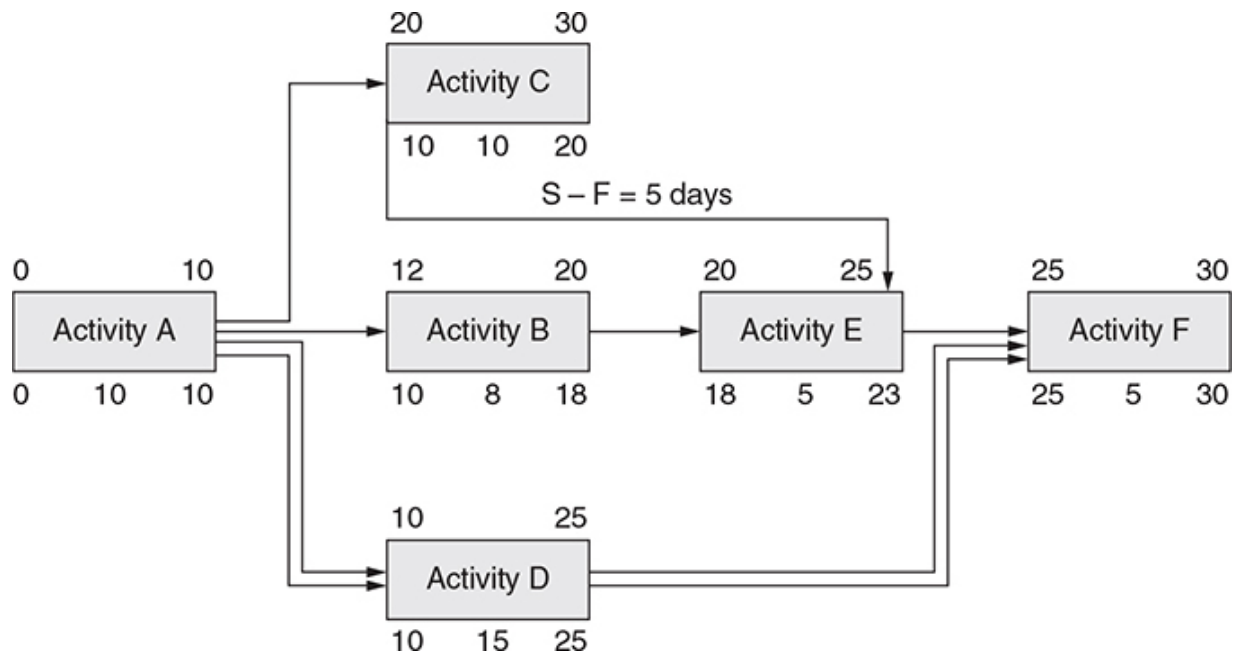
$$\begin{aligned}
 FF_C &= ES_E - ES_C - \text{Lag} \\
 &= 20 - 10 - 10 \\
 &= 0 \text{ days}
 \end{aligned}$$

Activity	Duration	Early start	Early finish	Late start	Late finish	Total float	Free float
A*	10	0	10	0	10	0	0
B	8	10	18	10	20	2	2
C*	10	10	20	10	20	0	0
D*	15	10	25	12	25	0	0
E*	5	20	25	20	25	0	0
F*	5	25	30	25	30	0	0

*Denotes critical activities.

Based on the forward and backward pass calculations, the critical path is through Activities A, C, D, E, and F. Note the free float of Activity C is governed by the S-S relationship between Activities C and E.

Example 10-6: Start-to-Finish Activity Relationships Although the start-to-finish relationship is seldom used, this example is presented to illustrate the calculations for an S-F relationship with a lag of 7 days between Activities C and E. The following diagram and accompanying table show the early and late start, early and late finish, and the total and free float for each activity.



LS	LF
Activity Description	
ES	EF

For Activity C, the calculation for total and free float are:

$$\begin{aligned}
 TF_C &= LF_C - EF_C \\
 &= 30 - 20 \\
 &= 10 \text{ days}
 \end{aligned}$$

$$\begin{aligned}
 FF_C &= EF_E - ES_C - \text{Lag} \\
 &= 23 - 10 - 5 \\
 &= 8 \text{ days}
 \end{aligned}$$

Activity	Duration	Early start	Early finish	Late start	Late finish	Total float	Free float
A*	10	0	10	0	10	0	0
B	8	10	18	12	20	2	0
C	10	10	20	20	30	10	8
D*	15	10	25	10	25	0	0
E	5	18	23	20	25	2	2
F*	5	25	30	25	30	0	0

*Denotes critical activities.

Based on the forward- and backward-pass calculations, the critical path is through Activities A, D, and F.

References